



Russound Controllers **(ACA, CAV, CAM, CAA, and CAS)**

RNET™ Protocol & Specifications **RS-232 Communication**

Document version 1.00.4

Quick Reference Guide

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1 Introduction

This document describes how to control and interpret data from Russound Controllers. Remote access is provided through the RS-232 serial port located on the rear panel of the Controllers (CAV also has front panel access) using the following **RNET™** Communications Protocol. It is assumed that the reader is familiar with the features and operation of the Controller being used. All commands use Hexadecimal or HEX values. In descriptive text these are indicated by preceding the value with a **0x**. Zero-based values are also used in certain areas and are noted as such. This means that a value of **0x00 = 1, 0x01 = 2, and 0x02 = 3** etc... Throughout this document all bytes within message packets not in bold must be referenced exactly as they appear.

2 Overview

Russound Controllers use a comprehensive Communications Protocol called **RNET™** which has been extended to the RS-232 port. Through this port, virtually all aspects of the device operation can be performed. This document provides detailed descriptions of messages required to perform basic device operation, as well as instructions on how to interpret Display Feedback messages.

When processing incoming 3rd Party requests, the ACA-E5 uses the 'Source Device ID' (see section 4.1.3 for more details on the Source Device ID) to determine which controller and zone is being accessed. Throughout the document, the Source Device ID field has been updated to reflect this (new) requirement.

In order to gain access to zone-specific data using the Set and Get messages (specified throughout section 7), it is necessary to issue at least one 'Event' message (see section 4.2.2 for more details on Event messages). This may be any of the existing Event messages or the new 'Device Restart' Event message (see section 5).

3 COM Port

The following information is needed to allow connection and communication with **RNET™** Controllers.

3.1 COM Settings

The following settings must be made to the RS-232 serial COM port used to interface with the Controllers:

- 19200 baud rate
- 1 Start bit
- 1 Stop bit
- No flow control
- No parity

3.2 Connector Type / Pin-out

Connector: Female DB-9

- Pin 1: NC
- Pin 2: Transmit
- Pin 3: Receive
- Pin 4: NC
- Pin 5: Ground
- Pin 6: NC
- Pin 7: NC
- Pin 8: NC
- Pin 9: NC

4 RNET™ Message Packet Format

Header				Body	Trailer	
0xF0	Target Device ID	Source Device ID	Type	Body	Checksum	0xF7

Every **RNET™** message has a consistent format. There are three major components: Message Header, Message Body and Message Trailer. Each of these is explained here in some detail.

MSB stands for **M**ost **S**ignificant **B**it which is the far left Bit of an 8-Bit Byte (MSB = Bit 7 = **00010001**). There are several special characters used in this protocol. These special characters have the MSB set to high (1). All Bytes within the message of an **RNET™** packet have the MSB set to low (0) except for three special characters. The special characters with the MSB set to high (1) are the “**Start of Message Character**”, the “**End of Message Character**”, and the “**Invert Character**”, which are explained within this document.

4.1 Message Header

Each message consists of a “Message Header” which is compiled of a **Start of Message Character**, **Target Device ID**, **Source Device ID**, and the **Message Type**.

4.1.1 Start of Message Character

As previously stated, there are several special characters. One of these is the Start of Message Character **0xF0** which is a special HEX value that indicates the beginning of a new message.

4.1.2 Target Device ID

The Target Device ID defines to which device the message is being sent. Every device on an **RNET™** system has a unique “Device ID” that allows messages to be sent to it. Each Device ID consists of a Controller ID, a Zone (Port) ID, and a Keypad ID. In the case of a Controller with zone Keypads, the purpose of each Device ID field is apparent.

4.1.2.1 Target Controller ID

On power up, the Controller assigns its Controller ID as well as a Zone ID and Keypad ID to all Keypads connected to it. The Controller ID is set in the Controllers “Setup Menu” using standard numbering (1 = 1, 2 = 2, etc...) but in the Device ID it is Zero-based (0x00 = 1, 0x01 = 2, 0x02 = 3, etc...). All systems have a controller 0x00 (the Root Controller) with additional controllers added (and numbered) as needed. A value of **0x7D** can be used to send messages to all Devices in the system.

The Russound **RNET™** system provides support for up to six connected Controllers for a maximum of 36 separate Zones. When a multiple Controller system is configured, each Controller is given a unique Controller ID through the programming procedure. All of the Controllers are linked together onto a common **RNET™** bus, so any messages sent to the serial port of one Controller will be available to all Controllers and all Keypads in the system.

All of the messages described in this document can be sent to any of the Controllers in the system by simply changing the Target Controller ID to match the Controller you would like to send the message to. There are several special Controller ID values that have been defined:

Here is a list of special Controller ID values:

Value (Hex)	Description
7F	All Keypads
7E	All Controllers
7D	All Devices

4.1.2.2 Target Zone (Port) ID

Zone ID defines the Zone for a particular device. For keypads, this defines which zone port on the Controllers rear panel that the keypad is connected to. The following special Zone ID values have been defined:

Here is a list of special Zone (Port) ID values:

Value (Hex)	Description
7F	Reserved
7E	Controller Link
7D	Peripheral Device (Internal Tuner, ST2-XM, etc...)
7C	Trace

Aside from the above list of special values and those for the actual ports on the back of a Controller (4-Zone Controllers = **0x00** – **0x03**) (6-Zone Controllers = **0x00** – **0x05**), any value can be used for the Zone ID field.

4.1.2.3 Target Keypad ID

Keypad ID identifies particular Keypads sharing a single Zone on the Controller or the Source Number of an Internal Tuner (AM/FM or XM for CAM only) when the Zone ID is set as Peripheral. **NOTE:** The CAM6.6 will only allow the Internal Tuner to be set as Source 1 and therefore must use a value of **0x00** in this field. The following values have specific meaning in the system:

Here is a list of special Keypad ID values:

Value (Hex)	Description
7F	The Controller Itself
7E	Reserved
7D	Targets all Keypads on a particular zone or all Sources for connected Peripheral Devices
7C	Used by Keypads to request an ID from the Controller
79	Indicates a Source Broadcast Display Feedback message

Aside from the above list of special values, all other Keypad IDs have not been formally assigned so they can be used as needed, any value can be used for the Zone ID field.

4.1.2.4 Target Device ID Example

Example 1 shows a complete Target Device ID for a Controller that has the Controller ID set to 1 and a Keypad plugged into Zone 1.

Example 1:

Value (HEX)	Device ID Field	Description
00	Controller ID	Controller 1 = 0x00 (Zero-based, see 4.1.2.1)
00	Zone (Port) ID	Zone 1 = 0x00 (Zero-based, see 4.1.2.2)
7F	Keypad ID	The Controller Itself

Example 2 shows a complete Target Device ID for a message sent to Zone 1 of all Controllers in the system such as All On / All Off.

Example 2:

Value (HEX)	Device ID Field	Description
7E	Controller ID	All Controllers
00	Zone (Port) ID	Zone 1 = 0x00 (Zero-based, see 4.1.2.2)
7F	Keypad ID	The Controller Itself

4.1.3 Source Device ID

The “Source” Device ID is the identification of the device that is sending the message. For external devices controlling the system, like an automation and control system, these can be any value that is a unique value among devices attached to the system. The recommend Device ID for external control systems is **Controller ID: 0x00, Zone ID: 0x00, and Keypad ID: 0x70 (0x71 when connected to an ACA-E5).**

4.1.3.1 Source Controller ID

For 3rd party devices, this should be set to a value of ‘0x00’.

4.1.3.2 Source Zone (Port) ID

For 3rd party devices, this should be set to a unique value among devices attached to the system.

4.1.3.3 Source Keypad ID

The recommend Keypad ID for external control systems is **0x70 (0x71 when connected to an ACA-E5)**, which is a Keypad ID other than those used in the system.

4.1.4 Message Type

This value defines the type of message that is being sent. The most important Message Types are as follows:

Value (Hex)	Message Type	Description
00	Set Data	Sets a parameter's value
01	Request Data	Requests a parameter's value
02	Handshake	Acknowledges a data send
05	Event	Triggers a system response that may set a parameter value, update displays, etc...

In some cases, setting parameter values can be done in two ways. A **Set Data** message can be sent directly to the parameter, or an **Event** message can be sent to trigger the Controller to set the value instead. **Event** messages are a little easier to use, and may trigger other desired results (such as updating Keypad displays, updating related parameters, etc...) where a **Set Data** message may not. Because of this, this document describes using **Event** messages to set parameter values where it is most desirable.

4.1.4.1 Paths

The Data within each RNET device is organized in a hierarchical tree similar to a computer's disk drive. When sending Data or Events from one system to another, a Path is specified to define exactly which object should be modified. Target and Source Paths are usually defined, but for the sake of remotely controlling the system, only the Target Path is of interest.

Paths in RNET messages are defined by first specifying how deep the path is, followed by the numeric values of each directory. For example, a Path that was three levels deep might be represented as "**3.1.2.8**". The "**3**" at the beginning defines how deep the Path is. The numbers that follow are the subdirectories. Think of the numbers as folder names (folder "**8**" is inside folder "**2**" which is inside folder "**1**"). Bear in mind that the folders are unique for each level, so folder "**1**" can be in folder "**1**" of the next level up (just like folders on a computer).

4.2 Message Body

The message body contains specific data which varies in value and byte count depending on the particular **Message Type** being sent. Refer to the particular messages.

4.2.1 The Invert Character

The **Invert Character** is used in special cases as part of the Message Body. If the data in an **RNET™** message body includes any byte values that have the MSB set to 1 (they have a hex value greater than **0x7F**), the byte will be rejected as only the lower 7 bits are used to hold data. In order to allow values greater than **0x7F** to be accepted, the byte must first be bitwise inverted (e.g., 10010101 = 01101010), and the special **Invert Character (0xF1)** is inserted just prior to the inverted byte. When an **RNET™** packet is received, the system must detect the **0xF1** invert

character. The **0xF1** character is then discarded and the following byte is inverted back to its original value (e.g., 01101010 = 10010101).

Invert Character Usage Example:

Sent message with inverted character:

Value (Hex)	Notes
F0	Start of message character
00	
67	
7C	
F1	Special Invert Character
6A	Inverted Character (actual value 0x95)
34	Checksum
F7	End of message character

Received message after re-inverting character:

Value (Hex)	Notes
F0	Start of message character
00	
67	
7C	
95	After re-inversion
XX	Checksum (Discarded)
F7	End of message character

4.2.2 Event Messages

Event Messages trigger a system response that may set a parameter value, update displays, etc...

Event Messages are a little easier to use than Data Messages and may trigger other desired results (such as updating Keypad displays, updating related parameters, etc...) where a **Set Data** message may not. Because of this, this document describes using **Event Messages** to set parameter values where it is most desirable. An **Event Message** Type consists of an Event ID, Event Timestamp, Event Data, and the Event Priority.

4.2.3 Data Messages

A **Set Data** message can be sent directly to the parameter to set a parameter's value.

4.3 Message Trailer

The Message Trailer consists of the **Checksum** and **End of Message Character**. The **Checksum** value changes and needs to be re-calculated whenever any one of the preceding characters in the message is changed. The **Checksum** is always followed by the **End of Message Character**.

4.3.1 Checksum

All messages include a **Checksum** that helps protect the integrity of the message. The **Checksum** is a single byte that can be calculated using the following formula (see example below):

Checksum Calculation Example:

Value (Hex)	Notes
F0	Start of Message Character
00	
67	
7C	
F1	
0F	
59	Checksum
F7	End of Message Character

Step #1 - Add the HEX value of every byte in the message that precedes the Checksum:

Example - $0xF0 + 0x00 + 0x67 + 0x7C + 0xF1 + 0x0F = 0x02D3$

Step #2 - Count the number of bytes which precede the Checksum and convert that value from DEC to HEX (byte count). Add the byte count in HEX to the previously calculated sum of bytes:

Example - $0x02D3 + 6$ (6 = Decimal value byte count) = $0x02D9$

Step #3 - This value is then **AND**-ed with the HEX value 0x007F (7F is the highest BIN value for 7 bits = 1111111). The **Checksum** itself and the **End of Message Character** are not included in the calculation. Only the low 7 bits are used so overflow is discarded:

Example - $0x02D9 \text{ AND } 0x007F = 0x59 = \text{Checksum}$

4.3.2 End of Message Character

The **End of Message Character (0xF7)** is the special HEX value that indicates the end of the message.

5 Device Restart Event

To indicate to the ACA-E5 controller that a Third Party device was added to the system, transmit a 'Device Restart Event message' as presented here. As mentioned previously, any other Event message will serve to indicate the addition of a Third Party device. In other words, a Device Restart Event is optional.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Value	F0	00	00	7F	cc	zz	kk	05	02	02	00	00	55	01	03	00	35	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = 0x71

xx = checksum

6 Buttons and Keycodes

This section describes how to send various events associated with the **Buttons** found on the keypads and/or the IR remote

The method for transmitting these button events falls into two categories:

-Keypad Events: The Keypad Events are the events that the keypad is capable of sending.

-Source Control Events: All others (Source Control Events) are only available via the IR Remote.

Note: all values are in **HEX**.

6.1 Keypad Events

Here is a list of UNO Keypad Event values and a description of which buttons they apply to:

Value (Hex)	Description
64	Setup Button
67	Previous
68	Next
69	Plus
6A	Minus
6B	Source (source toggle button)
6C	Power
6D	Stop
6E	Pause
6F	Favorite 1
70	Favorite 2
73	Play

7F	Volume Up
80	Volume Down (because this value is > 7F, the special F1 character must be used)

Here are some examples of some Keypad Events:

Zone 1, Volume Up

F0 00 00 7F 00 00 70 05 02 02 00 00 7F 00 00 00 00 01 7B F7
--

Zone 2, Play

F0 00 00 7F 00 01 70 05 02 02 00 00 73 00 00 00 00 01 70 F7
--

Zone 3, Favorite 1

F0 00 00 7F 00 02 70 05 02 02 00 00 6F 00 00 00 00 01 6D F7
--

Here is a break out of the **Zone1, Volume Up** message (using above example of Keypad Event)

Value	Field	Description
F0	Start of Message	
00	Target Controller ID	Controller 1
00	Target Zone ID	
7F	Target Keypad ID	The Controller itself
00	Source Controller ID	
00	Source Zone ID	Zone affected
70	Source Keypad ID	Arbitrary Keypad ID (not otherwise used by the system)
05	Message Type	Event Message
02	Target Path, Num Levels	
02	Target Path, Level 1	Root Menu
00	Target Path, Level 2	Run Menu
00	Source Path, Num Levels	No Source Path is used
7F	Event ID Lo Byte	Volume Up
00	Event ID Hi Byte	
00	Event Timestamp Lo Byte	unused
00	Event Timestamp Hi Byte	unused
00	Event Data Lo Byte	unused
00	Event Data Hi Byte	unused
01	Event Priority	Low Priority (does not generate a handshake)
7B	Checksum	Recalculate when a preceding byte value is changed
F7	End of Message	

6.2 Source Control Events

Source Control Events are unique to the System Remote and are all sent using the special “Remote Control Key” Event type with the actual Keycode passed in as the Data for the Event.

Here is a list of the Source Control Events (Keycodes):

Value (Hex)	Description
01	“1” Button
02	“2” Button
03	“3” Button
04	“4” Button
05	“5” Button
06	“6” Button
07	“7” Button
08	“8” Button
09	“9” Button
0A	“0” Button
0B	Volume Up
0C	Volume Down
0D	Mute (for zone, not source)
0E	Channel Up
0F	Channel Down
10	Power
11	Enter
12	Previous Channel
13	TV/Video
14	TV/VCR
15	A/B
16	TV/DVD
17	TV/LD
18	Input
19	TV/DSS
1A	Play
1B	Stop
1C	Search Forward
1D	Search Rewind
1E	Pause
1F	Record
20	Menu
21	Menu Up

22	Menu Down
23	Menu Left
24	Menu Right
25	Select
26	Exit
27	Display
28	Guide
29	Page Up
2A	Page Down
2B	Disk
2C	Plus 10
2D	Open/Close
2E	Random
2F	Track Forward
30	Track Reverse
31	Surround On/Off
32	Surround Mode
33	Surround Up
34	Surround Down
35	PIP
36	PIP Move
37	PIP Swap
38	Program
39	Sleep
3A	On
3B	Off
3C	11
3D	12
3E	13
3F	14
40	15
41	16
42	Bright
43	Dim
44	Close
45	Open
46	Stop 2
47	AM/FM

48	Cue
49	Disk Up
4A	Disk Down
4B	Info

Here are some examples of some Source Control Events:

Zone 1, Menu

F0 00 00 7F 00 00 70 05 02 02 00 00 F1 40 00 00 00 20 00 01 4E F7

Zone 2, Mute

F0 00 00 7F 00 01 70 05 02 02 00 00 F1 40 00 00 00 0D 00 01 3C F7

Zone 3, Record

F0 00 00 7F 00 02 70 05 02 02 00 00 F1 40 00 00 00 1F 00 01 49 F7

Here is a break out of the **Zone1, Menu** message used in the above example:

Value	Field	Description
F0	Start of Message	
00	Target Controller ID	Controller 1
00	Target Zone ID	
7F	Target Keypad ID	The Controller itself
00	Source Controller ID	
00	Source Zone ID	Zone affected
70	Source Keypad ID	Arbitrary Keypad ID (not otherwise used by the system)
05	Message Type	Event Message
02	Target Path, Num Levels	
02	Target Path, Level 1	Root Menu
00	Target Path, Level 2	Run Menu
00	Source Path, Num Levels	No Source Path is used
F1	Invert	Invert the next byte
40	Event ID Lo Byte	40 = 0xBF inverted = Remote Control Key Release
00	Event ID Hi Byte	
00	Event Timestamp Lo Byte	unused
00	Event Timestamp Hi Byte	unused
20	Event Data Lo Byte	Menu (Keycode)
00	Event Data Hi Byte	unused
01	Event Priority	Low Priority (does not generate a handshake)
7B	Checksum	Recalculate when a preceding

		byte value is changed
F7	End of Message	

7 Using Request Messages

The Request Data message is used to receive parameter data from the Controller. This may be used to receive Zone Power State, Volume Level, etc. When a Request Data message is sent to the Controller, a Set Data message is generated by the Controller and sent back to the Request Message sender. Since the Set Data message is of high priority, an Acknowledge message must be sent back to the Controller to acknowledge it received the Request message. Failure to send the Acknowledge message will result in a system delay of approximately 2.5 seconds. This is due to the Controller trying to re-send the Data.

Data can be requested discretely for each zone parameter, or all of the Zone information can be requested in a single message.

8 Zones

This section will provide information in regards to messages that only apply to Zone commands. This will include controlling the Zones as well as sending Request messages to “Get” the State of the Zones. Zero-based values are used for the Zone Numbers (i.e. **0 = 1, 1 = 2, and 2 = 3** etc...)

8.1 ON/OFF

The simplest way to explicitly turn a Zone On or turn a Zone Off is by using the discrete Zone On/Off Event. The Event Data fields determine the Zone and On/Off state. This example shows using the Zone On/Off Event message to execute a Zone On command for Zone 1 of Controller 1. Refer to the Buttons section for information on toggling zone On/Off (“Power” command found under Keypad Events).

8.1.1 Set State

Turn a specific Zone ON or OFF using a discrete message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Value	F0	cc	00	7F	cc	zz	kk	05	02	02	00	00	F1	23	00	##	00	zz	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #16 = 0x00 (off) or 0x01(on)

8.1.2 Get State

This is the **Request** message for the On/Off State of the selected Zone.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Value	F0	cc	00	7F	cc	zz	kk	01	04	02	00	zz	06	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Value	F0	cc	zz	kk	cc	00	7F	00	00	04	02	00	zz	06	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #21 = 0x00 (off) or 0x01(on)

8.1.3 Set All Zones On/Off State

The **RNET™** system can be sent a single message to issue an “All On” or “All Off” Event. The Controllers have the ability to enable or disable the All On/All Off state control per zone. See the Product Manual for instructions on programming each zone for “System On Enable” or “System On Disable”.

Turn All Zones ON or OFF using a discrete message (*all zones enabled in programming*).

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Value	F0	7E	00	7F	cc	zz	kk	05	02	02	00	00	F1	22	00	##	00	00	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #16 = 0x00 (all off) or 0x01(all on)

8.2 Source Select

The simplest way to explicitly select a Source per Zone is by using the discrete Source Select Event. The Event Data fields determine the Source being selected. Refer to the Buttons section for information on stepping through Sources (“*Source Toggle*” command found under *Keypad Events*). Zero-based values are used for the Source Numbers (i.e. **0 = 1, 1 = 2, and 2 = 3** etc...)

8.2.1 Set Source

Select the Source for a particular zone using a discrete message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Value	F0	cc	00	7F	cc	zz	kk	05	02	00	00	00	F1	3E	00	00	00	##	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #18 = selected source number -1

8.2.2 Get Source

This is the **Request** message for what Source is selected on a particular Zone.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Value	F0	cc	00	7F	cc	zz	kk	01	04	02	00	zz	02	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Value	F0	cc	zz	kk	cc	00	7F	00	00	04	02	00	zz	02	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #21 = source number -1

8.3 Volume Select

The simplest way to explicitly select a Volume level per Zone is by using the discrete Volume Select Event. The Even Timestamp fields determine the Volume Level. The Event Data fields determine the Zone affected by the Volume being selected. Refer to the Buttons section for information on stepping through Volume levels (“*Volume Up*” and “*Volume Down*” commands found under UNO Keypad Events). Volume levels displayed on UNO keypads range in value from 0 – 100 in steps of 2 (i.e. 0, 2, 4, 6, etc...). These Volume levels are represented in Decimal when using RS-232 messages ranging in value from 0 – 50 (i.e. 0 = 0, 1 = 2, ... 2 = 4). In the actual Data message the HEX value is used which would range in value from 0x00 – 0x32. This means (0x00 = 0 = 0, 0x01 = 1 = 2, 0x02 = 2 = 4 ... 0x32 = 50 = 100)

8.3.1 Set Volume

Select the Volume for a particular zone using a discrete message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Value	F0	cc	00	7F	cc	zz	kk	05	02	02	00	00	F1	21	00	##	00	zz	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #16 = volume level (0x00 - 0x32, 0x00 = 0 Displayed ... 0x32 = 100 Displayed)

8.3.2 Get Volume

This is the **Request** message for what Volume level is selected on a particular Zone.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Value	F0	cc	00	7F	cc	zz	kk	01	04	02	00	zz	01	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Value	F0	cc	zz	kk	cc	00	7F	00	00	04	02	00	zz	01	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #21 = Volume Level -1

8.4 Bass

Bass levels displayed on keypads range in value from -10 through +10. These Bass levels are represented in Decimal when using RS-232 messages ranging in value from 0 - 20. In the actual Data message the HEX value is used which would range in value from 0x00 – 0x14. This means (0x00 = 0 = -10, 0x01 = 1 = -9, 0x02 = 2 = -8 ... 0x14 = 20 = +10).

NOTE: The keypad displays will not automatically update for this change.

8.4.1 Bass Up/Bass Down

The following example shows the Bass level being increased or decreased.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	00	7F	cc	zz	kk	05	05	02	00	zz	00	00	00	##	00	00	00	01	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #16 = button event (Plus = 0x69 = Increase, Minus = 0x6A = Decrease)

8.4.2 Set Bass

Select the Bass level for a particular zone using a discrete message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	00	7F	cc	zz	kk	00	05	02	00	zz	00	00	00	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Bass level (0x00 = -10 ... 0x0A = Flat ... 0x14 = +10)

8.4.3 Get Bass

The current Bass setting for a particular zone can be obtained by using the following message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Value	F0	cc	00	7F	cc	zz	kk	01	05	02	00	zz	00	00	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	zz	kk	cc	00	7F	00	00	05	02	00	zz	00	00	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Bass level (0x00 = -10 ... 0x0A = Flat ... 0x14 = +10)

8.5 Treble

Treble levels displayed on keypads range in value from -10 through +10. These Treble levels are represented in Decimal when using RS-232 messages ranging in value from 0 - 20. In the actual Data message the HEX value is used which would range in value from 0x00 – 0x14. This means (0x00 = 0 = -10, 0x01 = 1 = -9, 0x02 = 2 = -8 ... 0x14 = 20 = +10).

NOTE: The keypad displays will not automatically update for this change.

8.5.1 Treble Up/Treble Down

The following example shows the Treble level being increased or decreased.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	00	7F	cc	zz	kk	05	05	02	00	zz	00	01	00	##	00	00	00	02	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #16 = button event (Plus = 0x69 = Increase, Minus = 0x6A = Decrease)

8.5.2 Set Treble

Select the Treble level for a particular zone using a discrete message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	00	7F	cc	zz	kk	00	05	02	00	zz	00	01	00	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Treble level (0x00 = -10 ... 0x0A = Flat ... 0x14 = +10)

8.5.3 Get Treble

The current Treble setting for a particular zone can be obtained by using the following message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Value	F0	cc	00	7F	cc	zz	kk	01	05	02	00	zz	00	01	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	zz	kk	cc	00	7F	00	00	05	02	00	zz	00	01	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Treble level (0x00 = -10 ... 0x0A = Flat ... 0x14 = +10)

8.6 Loudness

Loudness is displayed on keypads as “On” or “Off”. Loudness can be toggled On or Off with a Plus or Minus command. There can also be a discrete On or Off command selected.

NOTE: *The keypad displays will not automatically update for this change.*

8.6.1 Loudness Toggle On/Off

The following example shows how to toggle the Loudness “On” and “Off” for a particular Zone.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	00	7F	cc	zz	kk	05	05	02	00	zz	00	02	00	##	00	00	00	03	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #16 = button event (PLUS = 0x69 = On/Off toggle) or (MINUS = 0x6A = On/Off toggle)

8.6.2 Set Loudness

Turn Loudness On or Off for a particular zone using a discrete message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	00	7F	cc	zz	kk	00	05	02	00	zz	00	02	00	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Loudness setting (0x00 = OFF, 0x01 = ON)

8.6.3 Get Loudness

The current Loudness setting for a particular zone can be obtained by using the following message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Value	F0	cc	00	7F	cc	zz	kk	01	05	02	00	zz	00	02	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	zz	kk	cc	00	7F	00	00	05	02	00	zz	00	02	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Loudness setting (0x00 = OFF, 0x01 = ON)

8.7 Balance

Balance levels displayed on keypads range in value from “Left 10” to “Center” to “Right 10”. These Balance levels are represented in Decimal when using RS-232 messages ranging in value from 0 - 20. In the actual Data message the HEX value is used which would range in value from 0x00 – 0x14. This means (**0x00 = 0 = Left 10 ... 0x0A = 10 = Center ... 0x14 = 20 = Right 10**).

NOTE: *The keypad displays will not automatically update for this change.*

8.7.1 Balance Left or Balance Right

The following example shows the Balance level being drawn More Left or More Right.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	00	7F	cc	zz	kk	05	05	02	00	zz	00	03	00	##	00	00	00	04	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #16 = button event (Plus = 0x69 = More Left, Minus = 0x6A = More Right)

8.7.2 Set Balance

Select the Balance level for a particular zone using a discrete message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	00	7F	cc	zz	kk	00	05	02	00	zz	00	03	00	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Balance level (0x00 = More Left ... 0x0A = Center ... 0x14 = More Right)

8.7.3 Get Balance

The current Balance setting for a particular zone can be obtained using the following message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Value	F0	cc	00	7F	cc	zz	kk	01	05	02	00	zz	00	03	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	zz	kk	cc	00	7F	00	00	05	02	00	zz	00	03	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Balance level (0x00 = More Left ... 0x0A = Center ... 0x14 = More Right)

8.8 Turn On Volume

Turn On Volume levels displayed on keypads range in value from 0 – 100 in steps of 2 (i.e. 0, 2, 4 ... 100). These Turn On Volume levels are represented in Decimal when using RS-232 messages ranging in value from 0 - 50. In the actual Data message the HEX value is used which would range in value from 0x00 – 0x32. This means (**0x00 = 0 = 0, 0x01 = 1 = 2, 0x02 = 2 = 4 ... 0x32 = 50 = 100**)

NOTE: The keypad displays will not automatically update for this change.

8.8.1 Increase or Decrease Turn On Volume

The following example shows the Turn On Volume level being increased or decreased.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	00	7F	cc	zz	kk	05	05	02	00	zz	00	04	00	##	00	00	00	05	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #16 = button event (Plus = 0x69 = More Left, Minus = 0x6A = More Right)

8.8.2 Set Turn On Volume

Select the Turn On Volume level for a particular zone using a discrete message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	00	7F	cc	zz	kk	00	05	02	00	zz	00	04	00	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Turn On Volume level (0x00 - 0x32, 0x00 = 0 ... 0x32 = 100)

8.8.3 Get Turn On Volume

The current Turn On Volume for a particular zone can be obtained using the following message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Value	F0	cc	00	7F	cc	zz	kk	01	05	02	00	zz	00	04	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	zz	kk	cc	00	7F	00	00	04	02	00	zz	01	00	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Turn On Volume level (0x00 - 0x32, 0x00 = 0 ... 0x32 = 100)

8.9 Background Color

Background Color is displayed on keypads as “Amber”, “Green”, or “Off”. The Background Color can be toggled through these selections with a Plus or Minus command. There can also be a discrete command selected for each choice.

NOTE: The keypad displays *WILL* automatically update for this selection.

8.9.1 Background Color Off/Amber/Green toggle

The following example shows how to toggle the Background Color for a particular Zone.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	00	00	7F	cc	zz	kk	05	05	02	00	zz	00	05	00	##	00	00	00	06	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #16 = button event (Plus = 0x69 = Off/Color toggle) or (Minus = 0x6A = Off/Color toggle)

8.9.2 Set Background Color

Select the Background Color for a particular zone using a discrete message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	00	00	7F	cc	zz	kk	00	05	02	00	zz	00	05	00	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Background Color (0x00 = Off, 0x01 = Amber, 0x02 = Green)

8.9.3 Get Background Color

The current Background Color for a particular zone can be obtained using the following message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Value	F0	cc	00	7F	cc	zz	kk	01	05	02	00	zz	00	05	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	zz	kk	cc	00	7F	00	00	05	02	00	zz	00	05	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Background Color (0x00 = Off, 0x01 = Amber, 0x02 = Green)

8.10 Do Not Disturb

Do Not Disturb is displayed on keypads as “On” or “Off”. DND can be toggled On or Off with a Plus or Minus command. There can also be a discrete On or Off command selected.

8.10.1 Do Not Disturb On/Off Toggle

The following example shows how to toggle Do Not Disturb “On” and “Off” for a particular Zone.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	00	00	7F	cc	zz	kk	05	05	02	00	zz	00	06	00	##	00	00	00	07	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #16 = button event (Plus = 0x69 = On/Off toggle) or (Minus = 0x6A = On/Off toggle)

8.10.2 Set Do Not Disturb

Turn DND On or Off for a particular zone using a discrete message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	00	00	7F	cc	zz	kk	00	05	02	00	zz	00	06	00	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Do Not Disturb setting (0x00 = OFF, 0x01 = ON)

8.10.3 Get Do Not Disturb

The current DND setting for a particular zone can be obtained by using the following message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Value	F0	cc	00	7F	cc	zz	kk	01	05	02	00	zz	00	06	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	zz	kk	cc	00	7F	00	00	05	02	00	zz	00	06	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

Byte #22 = Do Not Disturb setting (0x00 = OFF, 0x01 = ON)

8.11 Party Mode

Party Mode is displayed on keypads as “On”, “Off”, or “Master”. Party Mode can be toggled to “On” or “Master” with the Plus command. The Minus command will toggle Party Mode “Off”. There can also be a discrete On, Off, or Master command sent.

8.11.1 Party Mode On/Master/Off

NOTE: If Party Mode is Off, then “Plus” will issue a “Master” command. If a Zone is turned on while the rest of the system has Party Mode On and a different Zone is the Master, then “Plus” from a non-active Zone will issue an “On” command first and then a “Master” command second. If Party Mode is On and a different Zone is Master, then “Plus” from an active zone will issue a “Master” command. Minus will always issue an “Off” command.

The following example shows how to toggle Party Mode for a particular Zone.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	00	00	7F	cc	zz	kk	05	05	02	00	zz	00	07	00	##	00	00	00	08	00	01	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #16 = button event (Plus = 0x69 = On/Off toggle) or (Minus = 0x6A = On/Off toggle)

8.11.2 Set Party Mode

Select Party Mode “Master”, “On”, or “Off” for a particular zone using a discrete message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	00	00	7F	cc	zz	kk	00	05	02	00	zz	00	07	00	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Party Mode setting (0x00 = OFF, 0x01 = ON, 0x02 = Master)

8.11.3 Get Party Mode

The current Party Mode state for a particular zone can be obtained using the following message.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Value	F0	cc	00	7F	cc	zz	kk	01	05	02	00	zz	00	07	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	cc	zz	kk	cc	00	7F	00	00	05	02	00	zz	00	07	00	00	01	00	01	00	##	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

Byte #22 = Party Mode setting (0x00 = OFF, 0x01 = ON, 0x02 = Master)

8.12 Get All Zone info

As stated previously, a message can be used to request all of a particular Zone's parameter values at once. This can be very useful for updating panel displays. The following is an example of how to request Zone information for a particular Zone and what the return message would look like.

8.12.1 Get State

This is the **Request** message for the parameter values of the selected Zone.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Value	F0	cc	00	7F	cc	zz	kk	01	04	02	00	zz	07	00	00	xx	F7

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The return message would look like the following.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Value	F0	cc	zz	kk	cc	00	7F	00	00	04	02	00	zz	07	00	00	01	00	0C	00	##	##	##
Byte #	24	25	26	27	28	29	30	31	32	33	34												
Value	##	##	##	##	##	##	##	##	00	xx	F7												

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = checksum

The parameter values are depicted in bytes 21 – 31. These values will change depending on the state of the selected Zone. The above example shows the parameter values for a Zone configured and being used as follows:

Byte #21 = Current Zone On/Off state (0x00 = OFF or 0x01 = ON)

Byte #22 = Current Source selected -1

Byte #23 = Current Volume level (0x00 - 0x32, 0x00 = 0 Displayed ... 0x32 = 100 Displayed)

Byte #24 = Current Bass level (0x00 = -10 ... 0x0A = Flat ... 0x14 = +10)

Byte #25 = Current Treble level (0x00 = -10 ... 0x0A = Flat ... 0x14 = +10)

Byte #26 = Current Loudness (0x00 = OFF, 0x01 = ON)

Byte #27 = Current Balance level (0x00 = More Left ... 0x0A = Center ... 0x14 = More Right)

Byte #28 = Current System On state (0x00 = All Zones Off, 0x01 = Any Zone is On)

Byte #29 = Current Shared Source (0x00 = Not Shared 0x01 = Shared with another Zone)

Byte #30 = Current Party Mode state (0x00 = OFF, 0x01 = ON, 0x02 = Master)*

Byte #31 = Current Do Not Disturb state (0x00 = OFF, 0x01 = ON)*

***NOTE:** Unsupported Features, information not available for CAS44 and CAA66 controllers

9 Displaying a String

Since the keypad contains a text display, we've provided a message that will allow you to send text messages to All Keypads simultaneously or a Specific Keypad individually.

“Alignment” – When sending a message to display text on a Keypad the first Data byte sets your Alignment (**0x00** = Centered, **0x01** = Left justified).

“Flash Time” – Messages can be displayed constant or they can be flashed on the display for a brief time. When sending a message to display text on a Keypad the second and third Data bytes set the Flash Time. Flash Time defines how long the string is displayed. This is measured in **10ms** increments with a value of **0x00** being **constant**.

“Null character” – A value of **0x00** used as a Data byte will be a text character of Null. A Null text character is always used after the last text character in the display string. With a 12 character text message you would use a Null character as the 13th text character. A Null character is also used for each unused text character at the end of the text display message if the maximum number of Data bytes are not all used.

9.1 On All Keypads

The following example shows how to display a text message on All Keypads (or other devices with a text display) in the system.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	7F	00	00	cc	zz	kk	00	02	01	01	00	00	00	01	00	10	00	##	##	##	##	##	##
Byte #	25	26	27	28	29	30	31	32	33	34	35	36												
Value	##	##	##	##	##	##	##	##	##	##	xx	F7												

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = Checksum

Byte #19 = Alignment (0x00 = Centered, 0x01 = Left justified)

Byte #20 = Low Byte of Flash Time

Byte #21 = High Byte of Flash Time

Byte #22 – #34 = ASCII text characters (Section 11 ASCII Character Set)

9.2 On A Specific Keypad

The following example shows how to display a text message on a Specific Keypad in the system. This is accomplished by setting the Target Device ID for the particular keypad in question.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	tc	tz	tk	cc	zz	kk	00	02	01	01	00	00	00	01	00	10	00	##	##	##	##	##	##
Byte #	25	26	27	28	29	30	31	32	33	34	35	36												
Value	##	##	##	##	##	##	##	##	##	##	xx	F7												

tc = TargetController Number -1

tz = Target Zone Number -1

tk = Target Keypad Number -1

cc = controller number -1

zz = zone number -1

kk = keypad id = 0x70 (0x71 when connected to an ACA-E5).

xx = Checksum

Byte #19 = Alignment (0x00 = Centered, 0x01 = Left justified)

Byte #20 = Low Byte of Flash Time

Byte #21 = High Byte of Flash Time

Byte #22 – #34 = ASCII text characters (Section 11 ASCII Character Set)

10 Display Messages

The following section describes how to read Display Messages. These include Direct Display Feedback messages, Source Broadcast Display Feedback messages, and Multi-Field Broadcast Display Feedback messages.

NOTE: *In order to fully support the display capabilities, Direct Display feedback messages, Source Broadcast Feedback messages and Multi-Field Broadcast Messages, should be supported. (Multi-Field Broadcast messages are only used with XM.)*

10.1 Reading Direct Display Feedback

This section describes how to read Direct Display Feedback Messages. These Feedback messages are usually sent in direct response to a received command (e.g., If the current frequency of a Russound tuner is “102.7 MHz FM”, sending the “Frequency Up” command will trigger the Tuner to send a Display message back to the sender to update the frequency Display to "102.9 MHz FM"). The Direct Display Feedback message is sent directly to the Target Device ID of the message sender. The message can be displayed for a constant amount of time, or a "Flash" display with a specified length of time (Flash Time is in increments of 10ms).

NOTE: *It is possible that some display messages will include the special "Invert" control character (0xF1).*

NOTE: *Some of the other bytes within this message may vary. Only the ones necessary to interpret the message are highlighted.*

NOTE: *This message shows a text payload of 16 characters. Some devices may have a text payload of 12 characters.*

This is what the Direct Display Feedback message would look like using the above example.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	00	00	70	00	7D	00	00	02	01	01	02	01	01	00	00	01	00	14	00	01	00	00	31
Byte #	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42						
Value	30	32	2E	39	20	4D	48	7A	20	46	4D	00	00	00	00	00	7F	F7						

Byte #2 – #4 = Target Device ID (This message should be displayed if the target Device ID matches the Device ID of your device.)

Byte #19 = Overall Payload Size

Byte #22 = Flash Time low byte (Flash time is in 10ms increments, **0x00** = Constant)

Byte #23 = Flash Time high byte

Byte #24 – #40 = Text (“102.9 MHz FM” used in above example)

Byte #41 = Calculated Checksum

10.2 Reading Source Broadcast Display Feedback

This section describes how to read Source Broadcast Display Feedback messages. These Feedback messages are sent to update all devices monitoring a given Source's status. These messages may be sent as a direct result of a sent command or as a general update. The Display Feedback message is sent with the source number of the Source attached. The attached source number indicates which Source the update is intended. The message can be displayed for a constant amount of time, or a "Flash" display with a specified length of time (Flash Time is in increments of 10ms).

NOTE: *It is possible that some display messages will include the special "Invert" control character (0xF1).*

NOTE: *Some of the other bytes within this message may vary. Only the ones necessary to interpret the message are highlighted.*

NOTE: *This message shows a text payload of 16 characters. Some devices may have a text payload of 12 characters.*

This example shows a Source Broadcast Display Feedback message with text "102.9 MHz FM".

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	7D	00	79	00	7D	00	00	02	01	01	02	01	01	00	00	01	00	14	00	10	00	00	31
Byte #	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42						
Value	30	32	2E	39	20	4D	48	7A	20	46	4D	00	00	00	00	00	14	F7						

Byte #4 = Target Keypad ID (**NOTE:** A value of **0x79** in the Target Keypad ID field indicates that this message is a Source Broadcast Display Feedback message.)

Byte #19 = Overall Payload Size

Byte #21 = Message type and Source Number = (0x10 (Source Broadcast Display Type) bit-wise **OR**-ed with the source number (e.g. source 1 = 0x10, source 3 = 0x12))

Byte #22 = Flash Time low byte (Flash time is in 10ms increments, **0x00** = Constant)

Byte #23 = Flash Time high byte

Bytes #24 – #40 = Text ("102.9 MHz FM")

Byte #41 = Calculated Checksum

10.3 Reading Multi-Field Broadcast Display Feedback Messages

Multi-Field Broadcast Display Feedback messages are sent to update all devices monitoring the Source's status. These Feedback messages are sent to update all devices monitoring a given Source's status. These messages may be sent as a direct result of a sent command or as a general update. The Display Feedback message is sent with the source number of the Source attached. A "Field ID" is included with the message to indicate which item is being updated (see table below). The message can be displayed for a constant amount of time, or a "Flash" display with a specified length of time (Flash Time is in increments of 10ms).

This example shows a Multi-Field Broadcast Display Feedback message text "49: Fine Tuning".

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Value	F0	7D	00	79	00	7D	00	00	02	01	01	02	01	01	00	00	01	00	14	00	20	07	1C	34
Byte #	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42						
Value	39	3A	46	69	6E	65	54	75	6E	69	6E	67	00	00	00	00	09	F7						

Byte #4 = Target Keypad ID (**NOTE:** A value of **0x79** in the Target Keypad ID field indicates that this message is a Source Broadcast Display Feedback message.)

Byte #19 = Overall Payload Size

Byte #21 = Message type and Source Number = (0x20 (Multi-Field Display Type) bit-wise OR-ed with the source number (e.g., source 1 = 0x20, source 3 = 0x22.))

Byte #22 = Field ID bit and Flash Bit = (bits 0 – 6 = Field Id (i.e., 0x07 = 7 = Channel Name), Bit 7 – Ignore. Invert control character (0xF1) will be inserted before this byte if Bit 7 is set.

Bytes #24 – #40 = Text ("49: Fine Tuning")

Byte #41 = Calculated Checksum

11 Using the Acknowledge Message (Handshaking)

When the controller sends a Return message in response to a Request message, the sender of the Request message must send an Acknowledge message in response to the Return message.

Failure to send the Acknowledge message will result in a system delay of approximately 2.5 seconds while the controller tries to re-send the data. The controller will also send an

Acknowledge message in response to a Set Data message that it is sent. It is not necessary to process the incoming Acknowledge message from the controller.

NOTE: A Peripheral Source does not require Handshaking when the keypad ID used to send data is 0x70. If using a keypad ID of 0x70, and therefore no Handshaking, it is best to leave approximately 100ms between messages to ensure that all messages are processed correctly. If Handshaking is desired, it is recommended to use a keypad ID of 0x60. In this case, the Peripheral Source will send an Acknowledge message in response to any high priority event messages sent. Acknowledge messages can be (and normally are) used to trigger the release of any additional messages that may be in the sender's queue.

This example shows sending a Frequency Up (Tune Up) event to a Russound Tuner: In this case the event priority byte is set to **0x00** (high) and the sender Keypad ID is **0x60**. This means that the Internal Tuner will send an Acknowledge message in response.

Byte #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Value	F0	00	7D	00	00	00	60	05	02	01	00	02	01	00	2F	00	70	00	00	00	00	0C	F7

Byte #4 = Source Number - 1 (Zero-based)

Byte #7 = Sender Keypad ID - 0x60 (0x60 indicates that this message requires a handshake message if the event is high priority)

Byte #15 = Event ID (TUNE UP)

Byte #17 = Return Keypad Id - 0x70 (This is the keypad ID that the ST2 will send any resulting direct display message to. If this byte is set to 0x70, it will not require you to send a handshake message in response to the display message. If the original Sender keypad ID of 0x60 is used here, you will need to send a handshake message when the direct display message comes back. Failure to send the handshake message will result in a system delay of approximately 2.5 seconds, while the ST2 tries to re-send the data. It is recommended to leave this byte as 0x70.)

Byte #19 = Source Number - 1 (Zero-based)

Byte #21 = Event Priority (0 = high, 1 = low)

Byte #22 = Calculated Checksum

Handshake message returned to sender to acknowledge a high priority event message

Byte #	1	2	3	4	5	6	7	8	9	10	11
Value	F0	00	00	60	00	7D	00	02	06	5E	F7

Byte #7 = Source Number of Event Recipient - 1 (Zero-based)

Byte #8 = Message Type (HANDSHAKE)

Byte #9 = Handshake Type (EVENT HANDSHAKE)

Byte #10 = Calculated Checksum

12 ASCII Character Set to HEX Conversion Chart

ASCII	=	HEX	ASCII	=	HEX	ASCII	=	HEX
Space		20	@		40	`		60
!		21	A		41	a		61
"		22	B		42	b		62
#		23	C		43	c		63
\$		24	D		44	d		64
%		25	E		45	e		65
&		26	F		46	f		66
'		27	G		47	g		67
(		28	H		48	h		68
)		29	I		49	i		69
*		2A	J		4A	j		6A
+		2B	K		4B	k		6B
,		2C	L		4C	l		6C
-		2D	M		4D	m		6D
.		2E	N		4E	n		6E
/		2F	O		4F	o		6F
0		30	P		50	p		70
1		31	Q		51	q		71
2		32	R		52	r		72
3		33	S		53	s		73
4		34	T		54	t		74
5		35	U		55	u		75
6		36	V		56	v		76
7		37	W		57	w		77
8		38	X		58	x		78
9		39	Y		59	y		79
:		3A	Z		5A	z		7A
;		3B	[		5B	{		7B
<		3C	\		5C			7C
=		3D	]		5D	}		7D
>		3E	^		5E	~		7E
?		3F	_		5F	Del		7F

13 Russound Controllers RS-232 Hex Code Listing

Russound Controllers RS-232 Hex Code Listing																						
CONTROLLER #1 KEYPAD EVENTS - ZONE POWER ON AND OFF COMMANDS																						
	F0		00	7F	00	00	70	05	02	02	00	00	F1	23	00		00		00	01		F7
Byte Count	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
ZONE 1 ON	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	01	00	00	00	01	12	F7
ZONE 1 OFF	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	00	00	00	00	01	11	F7
ZONE 2 ON	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	01	00	01	00	01	13	F7
ZONE 2 OFF	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	00	00	01	00	01	12	F7
ZONE 3 ON	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	01	00	02	00	01	14	F7
ZONE 3 OFF	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	00	00	02	00	01	13	F7
ZONE 4 ON	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	01	00	03	00	01	15	F7
ZONE 4 OFF	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	00	00	03	00	01	14	F7
ZONE 5 ON	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	01	00	04	00	01	16	F7
ZONE 5 OFF	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	00	00	04	00	01	15	F7
ZONE 6 ON	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	01	00	05	00	01	17	F7
ZONE 6 OFF	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	23	00	00	00	05	00	01	16	F7
CONTROLLER #1 - ALL ON / ALL OFF																						
	F0	7E	00	7F	00	00	70	05	02	02	00	00	F1	22	00	00		00	00	01		F7
Byte Count	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
ALL ON	F0	7E	00	7F	00	00	70	05	02	02	00	00	F1	22	00	00	01	00	00	01	0F	F7
ALL OFF	F0	7E	00	7F	00	00	70	05	02	02	00	00	F1	22	00	00	00	00	00	01	0E	F7
CONTROLLER #1 KEYPAD EVENTS - ZONE VOLUME UP/DOWN COMMANDS																						
	F0	00	00	7F	00		70	05	02	02	00	00		00	00	00	00	00	01		F7	
Byte Count	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
ZONE 1 VOL UP	F0	00	00	7F	00	00	70	05	02	02	00	00	7F	00	00	00	00	00	01	7B	F7	
ZONE 1 VOL DN	F0	00	00	7F	00	00	70	05	02	02	00	00	F1	7F	00	00	00	00	00	01	6D	F7
ZONE 2 VOL UP	F0	00	00	7F	00	01	70	05	02	02	00	00	7F	00	00	00	00	00	01	7C	F7	
ZONE 2 VOL DN	F0	00	00	7F	00	01	70	05	02	02	00	00	F1	7F	00	00	00	00	00	01	6E	F7
ZONE 3 VOL UP	F0	00	00	7F	00	02	70	05	02	02	00	00	7F	00	00	00	00	00	01	7D	F7	
ZONE 3 VOL DN	F0	00	00	7F	00	02	70	05	02	02	00	00	F1	7F	00	00	00	00	00	01	6F	F7
ZONE 4 VOL UP	F0	00	00	7F	00	03	70	05	02	02	00	00	7F	00	00	00	00	00	01	7E	F7	
ZONE 4 VOL DN	F0	00	00	7F	00	03	70	05	02	02	00	00	F1	7F	00	00	00	00	00	01	70	F7
ZONE 5 VOL UP	F0	00	00	7F	00	04	70	05	02	02	00	00	7F	00	00	00	00	00	01	7F	F7	
ZONE 5 VOL DN	F0	00	00	7F	00	04	70	05	02	02	00	00	F1	7F	00	00	00	00	00	01	71	F7
ZONE 6 VOL UP	F0	00	00	7F	00	05	70	05	02	02	00	00	7F	00	00	00	00	00	01	00	F7	
ZONE 6 VOL DN	F0	00	00	7F	00	05	70	05	02	02	00	00	F1	7F	00	00	00	00	00	01	72	F7

CONTROLLER #1 KEYPAD EVENTS - ZONE PREVIOUS/NEXT & PLUS/MINUS COMMANDS																						
Byte Count	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	
ZONE 1 PREV	F0	00	00	7F	00	00	70	05	02	02	00	00	67	00	00	00	00	00	01	63	F7	
ZONE 1 NEXT	F0	00	00	7F	00	00	70	05	02	02	00	00	68	00	00	00	00	00	01	64	F7	
ZONE 1 PLUS	F0	00	00	7F	00	00	70	05	02	02	00	00	69	00	00	00	00	00	01	65	F7	
ZONE 1 MINUS	F0	00	00	7F	00	00	70	05	02	02	00	00	6A	00	00	00	00	00	01	66	F7	
ZONE 2 PREV	F0	00	00	7F	00	01	70	05	02	02	00	00	67	00	00	00	00	00	01	64	F7	
ZONE 2 NEXT	F0	00	00	7F	00	01	70	05	02	02	00	00	68	00	00	00	00	00	01	65	F7	
ZONE 2 PLUS	F0	00	00	7F	00	01	70	05	02	02	00	00	69	00	00	00	00	00	01	66	F7	
ZONE 2 MINUS	F0	00	00	7F	00	01	70	05	02	02	00	00	6A	00	00	00	00	00	01	67	F7	
ZONE 3 PREV	F0	00	00	7F	00	02	70	05	02	02	00	00	67	00	00	00	00	00	01	65	F7	
ZONE 3 NEXT	F0	00	00	7F	00	02	70	05	02	02	00	00	68	00	00	00	00	00	01	66	F7	
ZONE 3 PLUS	F0	00	00	7F	00	02	70	05	02	02	00	00	69	00	00	00	00	00	01	67	F7	
ZONE 3 MINUS	F0	00	00	7F	00	02	70	05	02	02	00	00	6A	00	00	00	00	00	01	68	F7	
ZONE 4 PREV	F0	00	00	7F	00	03	70	05	02	02	00	00	67	00	00	00	00	00	01	66	F7	
ZONE 4 NEXT	F0	00	00	7F	00	03	70	05	02	02	00	00	68	00	00	00	00	00	01	67	F7	
ZONE 4 PLUS	F0	00	00	7F	00	03	70	05	02	02	00	00	69	00	00	00	00	00	01	68	F7	
ZONE 4 MINUS	F0	00	00	7F	00	03	70	05	02	02	00	00	6A	00	00	00	00	00	01	69	F7	
ZONE 5 PREV	F0	00	00	7F	00	04	70	05	02	02	00	00	67	00	00	00	00	00	01	67	F7	
ZONE 5 NEXT	F0	00	00	7F	00	04	70	05	02	02	00	00	68	00	00	00	00	00	01	68	F7	
ZONE 5 PLUS	F0	00	00	7F	00	04	70	05	02	02	00	00	69	00	00	00	00	00	01	69	F7	
ZONE 5 MINUS	F0	00	00	7F	00	04	70	05	02	02	00	00	6A	00	00	00	00	00	01	6A	F7	
ZONE 6 PREV	F0	00	00	7F	00	05	70	05	02	02	00	00	67	00	00	00	00	00	01	68	F7	
ZONE 6 NEXT	F0	00	00	7F	00	05	70	05	02	02	00	00	68	00	00	00	00	00	01	69	F7	
ZONE 6 PLUS	F0	00	00	7F	00	05	70	05	02	02	00	00	69	00	00	00	00	00	01	6A	F7	
ZONE 6 MINUS	F0	00	00	7F	00	05	70	05	02	02	00	00	6A	00	00	00	00	00	01	6B	F7	

CONTROLLER #1 KEYPAD EVENTS - ZONE PLAY / STOP / PAUSE / F1 / F2 COMMANDS																									
			F0	00	00	7F	00		70	05	02	02	00	00		00	00	00	00	01		F7			
Byte Count			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21		
ZONE 1 PLAY			F0	00	00	7F	00	00	70	05	02	02	00	00	73	00	00	00	00	00	01	6F	F7		
ZONE 1 STOP			F0	00	00	7F	00	00	70	05	02	02	00	00	6D	00	00	00	00	00	01	69	F7		
ZONE 1 PAUSE			F0	00	00	7F	00	00	70	05	02	02	00	00	6E	00	00	00	00	00	01	6A	F7		
ZONE 1 F1			F0	00	00	7F	00	00	70	05	02	02	00	00	6F	00	00	00	00	00	01	6B	F7		
ZONE 1 F2			F0	00	00	7F	00	00	70	05	02	02	00	00	70	00	00	00	00	00	01	6C	F7		
ZONE 2 PLAY			F0	00	00	7F	00	01	70	05	02	02	00	00	73	00	00	00	00	00	01	70	F7		
ZONE 2 STOP			F0	00	00	7F	00	01	70	05	02	02	00	00	6D	00	00	00	00	00	01	6A	F7		
ZONE 2 PAUSE			F0	00	00	7F	00	01	70	05	02	02	00	00	6E	00	00	00	00	00	01	6B	F7		
ZONE 2 F1			F0	00	00	7F	00	01	70	05	02	02	00	00	6F	00	00	00	00	00	01	6C	F7		
ZONE 2 F2			F0	00	00	7F	00	01	70	05	02	02	00	00	70	00	00	00	00	00	01	6D	F7		
ZONE 3 PLAY			F0	00	00	7F	00	02	70	05	02	02	00	00	73	00	00	00	00	00	01	71	F7		
ZONE 3 STOP			F0	00	00	7F	00	02	70	05	02	02	00	00	6D	00	00	00	00	00	01	6B	F7		
ZONE 3 PAUSE			F0	00	00	7F	00	02	70	05	02	02	00	00	6E	00	00	00	00	00	01	6C	F7		
ZONE 3 F1			F0	00	00	7F	00	02	70	05	02	02	00	00	6F	00	00	00	00	00	01	6D	F7		
ZONE 3 F2			F0	00	00	7F	00	02	70	05	02	02	00	00	70	00	00	00	00	00	01	6E	F7		
ZONE 4 PLAY			F0	00	00	7F	00	03	70	05	02	02	00	00	73	00	00	00	00	00	01	72	F7		
ZONE 4 STOP			F0	00	00	7F	00	03	70	05	02	02	00	00	6D	00	00	00	00	00	01	6C	F7		
ZONE 4 PAUSE			F0	00	00	7F	00	03	70	05	02	02	00	00	6E	00	00	00	00	00	01	6D	F7		
ZONE 4 F1			F0	00	00	7F	00	03	70	05	02	02	00	00	6F	00	00	00	00	00	01	6E	F7		
ZONE 4 F2			F0	00	00	7F	00	03	70	05	02	02	00	00	70	00	00	00	00	00	01	6F	F7		
ZONE 5 PLAY			F0	00	00	7F	00	04	70	05	02	02	00	00	73	00	00	00	00	00	01	73	F7		
ZONE 5 STOP			F0	00	00	7F	00	04	70	05	02	02	00	00	6D	00	00	00	00	00	01	6D	F7		
ZONE 5 PAUSE			F0	00	00	7F	00	04	70	05	02	02	00	00	6E	00	00	00	00	00	01	6E	F7		
ZONE 5 F1			F0	00	00	7F	00	04	70	05	02	02	00	00	6F	00	00	00	00	00	01	6F	F7		
ZONE 5 F2			F0	00	00	7F	00	04	70	05	02	02	00	00	70	00	00	00	00	00	01	70	F7		
ZONE 6 PLAY			F0	00	00	7F	00	05	70	05	02	02	00	00	73	00	00	00	00	00	01	74	F7		
ZONE 6 STOP			F0	00	00	7F	00	05	70	05	02	02	00	00	6D	00	00	00	00	00	01	6E	F7		
ZONE 6 PAUSE			F0	00	00	7F	00	05	70	05	02	02	00	00	6E	00	00	00	00	00	01	6F	F7		
ZONE 6 F1			F0	00	00	7F	00	05	70	05	02	02	00	00	6F	00	00	00	00	00	01	70	F7		
ZONE 6 F2			F0	00	00	7F	00	05	70	05	02	02	00	00	70	00	00	00	00	00	01	71	F7		

CONTROLLER #1 KEYPAD EVENTS - ZONE SOURCE SELECT COMMANDS																								
			F0	00	00	7F	00		70	05	02	00	00	00	F1	3E	00	00	00		00	01		F7
Byte Count	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22		
ZONE 1 SRC 1	F0	00	00	7F	00	00	70	05	02	00	00	00	F1	3E	00	00	00	00	00	01	2A	F7		
ZONE 1 SRC 2	F0	00	00	7F	00	00	70	05	02	00	00	00	F1	3E	00	00	00	01	00	01	2B	F7		
ZONE 1 SRC 3	F0	00	00	7F	00	00	70	05	02	00	00	00	F1	3E	00	00	00	02	00	01	2C	F7		
ZONE 1 SRC 4	F0	00	00	7F	00	00	70	05	02	00	00	00	F1	3E	00	00	00	03	00	01	2D	F7		
ZONE 1 SRC 5	F0	00	00	7F	00	00	70	05	02	00	00	00	F1	3E	00	00	00	04	00	01	2E	F7		
ZONE 1 SRC 6	F0	00	00	7F	00	00	70	05	02	00	00	00	F1	3E	00	00	00	05	00	01	2F	F7		
ZONE 1 AUX	F0	00	00	7F	00	00	70	05	02	00	00	00	F1	3E	00	00	00	06	00	01	30	F7		
ZONE 2 SRC 1	F0	00	00	7F	00	01	70	05	02	00	00	00	F1	3E	00	00	00	00	00	01	2B	F7		
ZONE 2 SRC 2	F0	00	00	7F	00	01	70	05	02	00	00	00	F1	3E	00	00	00	01	00	01	2C	F7		
ZONE 2 SRC 3	F0	00	00	7F	00	01	70	05	02	00	00	00	F1	3E	00	00	00	02	00	01	2D	F7		
ZONE 2 SRC 4	F0	00	00	7F	00	01	70	05	02	00	00	00	F1	3E	00	00	00	03	00	01	2E	F7		
ZONE 2 SRC 5	F0	00	00	7F	00	01	70	05	02	00	00	00	F1	3E	00	00	00	04	00	01	2F	F7		
ZONE 2 SRC 6	F0	00	00	7F	00	01	70	05	02	00	00	00	F1	3E	00	00	00	05	00	01	30	F7		
ZONE 2 AUX	F0	00	00	7F	00	01	70	05	02	00	00	00	F1	3E	00	00	00	06	00	01	31	F7		
ZONE 3 SRC 1	F0	00	00	7F	00	02	70	05	02	00	00	00	F1	3E	00	00	00	00	00	01	2C	F7		
ZONE 3 SRC 2	F0	00	00	7F	00	02	70	05	02	00	00	00	F1	3E	00	00	00	01	00	01	2D	F7		
ZONE 3 SRC 3	F0	00	00	7F	00	02	70	05	02	00	00	00	F1	3E	00	00	00	02	00	01	2E	F7		
ZONE 3 SRC 4	F0	00	00	7F	00	02	70	05	02	00	00	00	F1	3E	00	00	00	03	00	01	2F	F7		
ZONE 3 SRC 5	F0	00	00	7F	00	02	70	05	02	00	00	00	F1	3E	00	00	00	04	00	01	30	F7		
ZONE 3 SRC 6	F0	00	00	7F	00	02	70	05	02	00	00	00	F1	3E	00	00	00	05	00	01	31	F7		
ZONE 3 AUX	F0	00	00	7F	00	02	70	05	02	00	00	00	F1	3E	00	00	00	06	00	01	32	F7		
ZONE 4 SRC 1	F0	00	00	7F	00	03	70	05	02	00	00	00	F1	3E	00	00	00	00	00	01	2D	F7		
ZONE 4 SRC 2	F0	00	00	7F	00	03	70	05	02	00	00	00	F1	3E	00	00	00	01	00	01	2E	F7		
ZONE 4 SRC 3	F0	00	00	7F	00	03	70	05	02	00	00	00	F1	3E	00	00	00	02	00	01	2F	F7		
ZONE 4 SRC 4	F0	00	00	7F	00	03	70	05	02	00	00	00	F1	3E	00	00	00	03	00	01	30	F7		
ZONE 4 SRC 5	F0	00	00	7F	00	03	70	05	02	00	00	00	F1	3E	00	00	00	04	00	01	31	F7		
ZONE 4 SRC 6	F0	00	00	7F	00	03	70	05	02	00	00	00	F1	3E	00	00	00	05	00	01	32	F7		
ZONE 4 AUX	F0	00	00	7F	00	03	70	05	02	00	00	00	F1	3E	00	00	00	06	00	01	33	F7		
ZONE 5 SRC 1	F0	00	00	7F	00	04	70	05	02	00	00	00	F1	3E	00	00	00	00	00	01	2E	F7		
ZONE 5 SRC 2	F0	00	00	7F	00	04	70	05	02	00	00	00	F1	3E	00	00	00	01	00	01	2F	F7		
ZONE 5 SRC 3	F0	00	00	7F	00	04	70	05	02	00	00	00	F1	3E	00	00	00	02	00	01	30	F7		
ZONE 5 SRC 4	F0	00	00	7F	00	04	70	05	02	00	00	00	F1	3E	00	00	00	03	00	01	31	F7		
ZONE 5 SRC 5	F0	00	00	7F	00	04	70	05	02	00	00	00	F1	3E	00	00	00	04	00	01	32	F7		
ZONE 5 SRC 6	F0	00	00	7F	00	04	70	05	02	00	00	00	F1	3E	00	00	00	05	00	01	33	F7		
ZONE 5 AUX	F0	00	00	7F	00	04	70	05	02	00	00	00	F1	3E	00	00	00	06	00	01	34	F7		
ZONE 6 SRC 1	F0	00	00	7F	00	05	70	05	02	00	00	00	F1	3E	00	00	00	00	00	01	2F	F7		
ZONE 6 SRC 2	F0	00	00	7F	00	05	70	05	02	00	00	00	F1	3E	00	00	00	01	00	01	30	F7		
ZONE 6 SRC 3	F0	00	00	7F	00	05	70	05	02	00	00	00	F1	3E	00	00	00	02	00	01	31	F7		
ZONE 6 SRC 4	F0	00	00	7F	00	05	70	05	02	00	00	00	F1	3E	00	00	00	03	00	01	32	F7		
ZONE 6 SRC 5	F0	00	00	7F	00	05	70	05	02	00	00	00	F1	3E	00	00	00	04	00	01	33	F7		
ZONE 6 SRC 6	F0	00	00	7F	00	05	70	05	02	00	00	00	F1	3E	00	00	00	05	00	01	34	F7		
ZONE 6 AUX	F0	00	00	7F	00	05	70	05	02	00	00	00	F1	3E	00	00	00	06	00	01	35	F7		

